

Trainings ▾

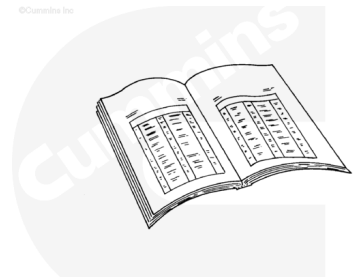
Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



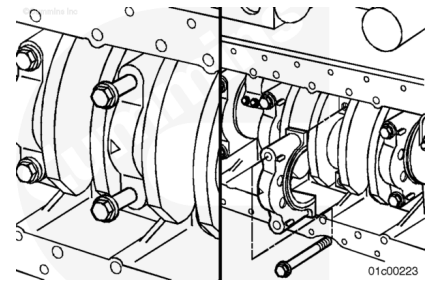
ck800wa

- Drain the lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 (Lubricating Oil Pan) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 (Lubricating Oil Suction Tube (Block Mounted)) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Remove the lubricating oil pump. Refer to Procedure 007-031 (Lubricating Oil Pump) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 (Block Stiffener Plate) in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)

Remove

⚠ CAUTION ⚠

Mark any unmarked bearing caps before removing them from the cylinder block. Do not remove main bearing capscrews until the main bearing cap is pulled, or the cap could fall out, causing personal injury and parts damage.



If the crankshaft is **not** going to be removed, replace the main bearings one at a time.

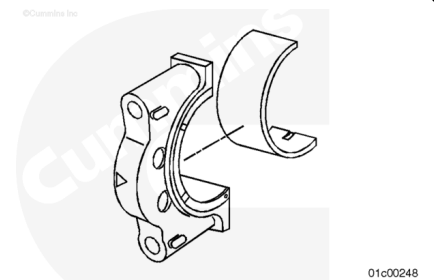
Loosen the main bearing cap capscrews.

Use the main bearing cap puller, Part Number ST-1178, to remove the cap. The tool **must** be centered on the cap.

Remove the main bearing capscrews and cap.

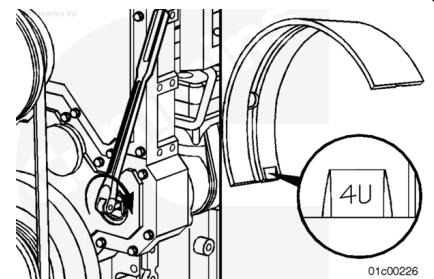
Mark the bearing shells with the journal number from which they were removed.

Remove the lower main bearing shell from the cap.



Rotate the crankshaft with the air compressor drive or a barring device to remove the main bearing shell.

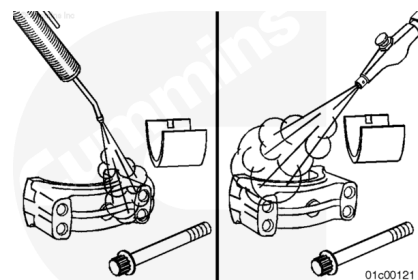
Mark each main bearing for location if they are to be installed again in the engine.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



Use a soft bristle brush to clean the parts.

Dry the parts with compressed air.

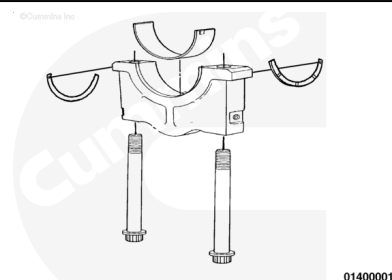
Inspect the main bearing cap and capscrews for damage.

Inspect the bearing shells and thrust bearings for nicks, scratches, or other damage.

If the main bearings are damaged, inspect the crankshaft main bearing journals. If the crankshaft is damaged, the engine **must** be removed for repair. Refer to Procedure 000-001 (Engine Removal) in Section 0.

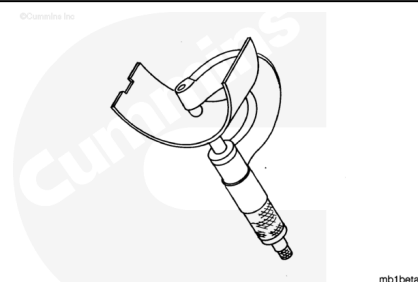
(/qs3/pubsys2/xml/en/procedures/10/10-000-001.html)

Cummins Inc. recommends replacing the thrust bearings when the main bearings are replaced. Refer to Procedure 001-007 (Bearings, Thrust) in Section 1 for thrust bearing replacement. (/qs3/pubsys2/xml/en/procedures/10/10-001-007-tr.html)



Use an outside diameter ball-tipped micrometer to measure the main bearing shell thickness.

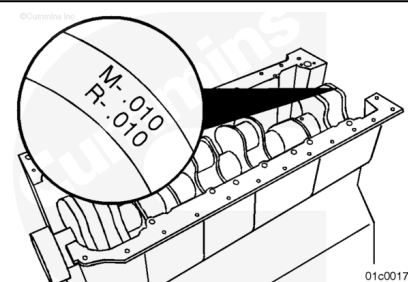
Standard Main Bearing Shell Thickness		
mm		in
3.196	MIN	0.126
3.214	MAX	0.127



If the bearing shell is **not** within specifications, it **must** be replaced.

For more detailed information on bearing damage, refer to Analysis and Prevention of Bearing Failures, Bulletin 3810387.

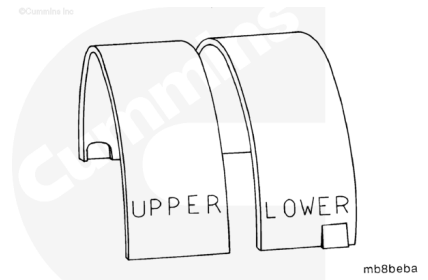
Crankshafts that are ground undersize in the connecting rod and main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.



Install

⚠ CAUTION ⚠

The cylinder block saddle and the cap mating surfaces must be clean and dry when the bearing shells are installed. Used bearings must be installed in their original location or engine damage can occur.



⚠ CAUTION ⚠

The bearing tang must fit into the slot in the bearing saddle or engine damage can occur.

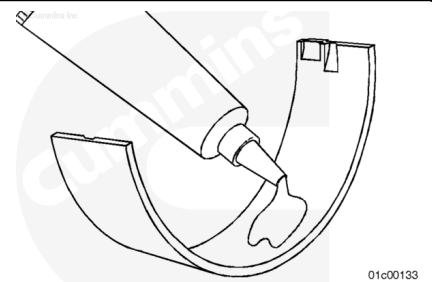
⚠ CAUTION ⚠

Do not use a capscrew or any hard metal object to install bearings, as this can damage the crankshaft and cause an engine failure.

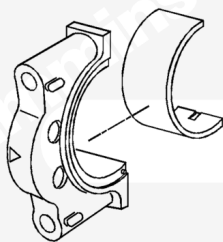
Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the upper main bearing shell to crankshaft journal mating surface.

Start the bearing into the upper saddle by hand. Slide the bearing into position. Take care to **not** damage the crankshaft journal while installing the bearing.

Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the inside diameter of the bearing shells and the liner side of the upper thrust bearings.



Align the lower main bearing shell and push it into place.



01c00227

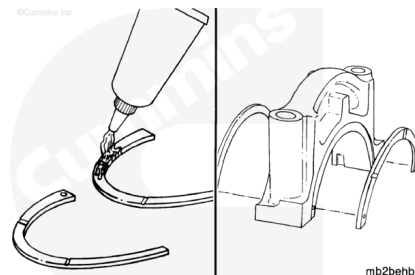
⚠ CAUTION ⚠

Make sure the bevel side is toward the crankshaft journal. Incorrect thrust bearing installation will result in engine damage.

Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the liner side of the lower thrust bearings.

Install the lower thrust bearings in the number four main bearing cap as shown.

© Cummins Inc.



mb2behb

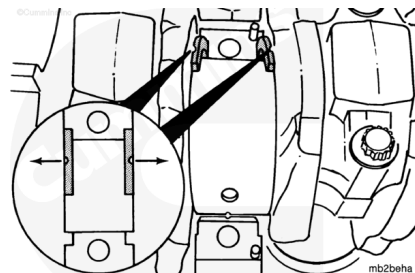
⚠ CAUTION ⚠

Make sure the bevel side is toward the crankshaft journal. Incorrect thrust bearing installation will result in engine damage.

Install the upper thrust bearings in the number four main bearing saddle.

Push the crankshaft toward the rear of the engine to install the front thrust bearing and toward the front of the engine to install the rear thrust bearing.

© Cummins Inc.

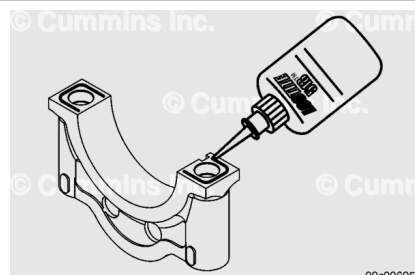


mb2beha

⚠ CAUTION ⚠

Only use Loctite™ 518 thread sealant on the main cap joint mating surface. Other sealants can become hard, brittle, and allow oil and debris into the main bearing/block joint, causing engine damage.

© Cummins Inc.



00c00605

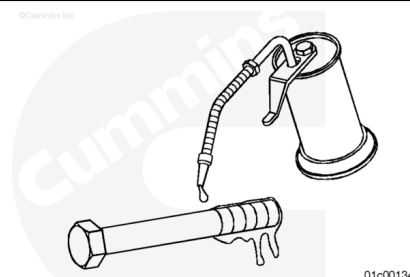
⚠ CAUTION ⚠

The bead must be 3 to 5 mm [0.12 to 0.20 in] wide and must not enter the main bearing shell inside the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.

Apply a bead of Loctite™ 518 thread sealant to the two main bearing cap surfaces on each main bearing cap as shown.

Use clean 15W-40 engine oil to lubricate the capscrew threads and flat washers (or the underside of the capscrew heads if washers are **not** used).

Drain the excess oil from the capscrews before installing them.

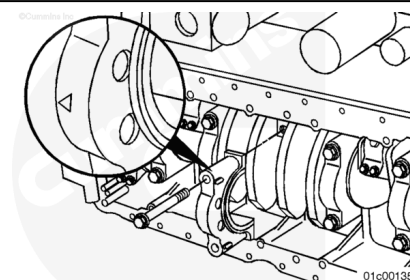


⚠ CAUTION ⚠

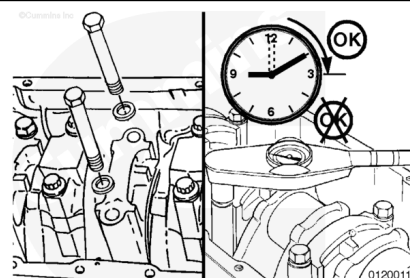
The main bearing caps are marked with a 'V'. Position the 'V' mark so that it points to the front of the engine. Incorrect positioning will result in severe engine damage.

Align the capscrew holes in the cap with the holes in the cylinder block. Make sure the lower bearing shell is in position.

Install the capscrews through the cap and into the cylinder block.

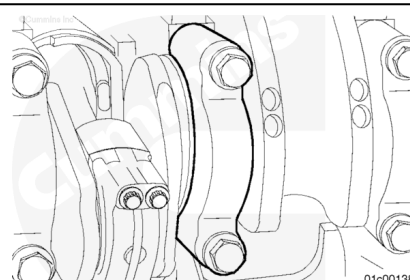


To reduce the possibility of bore size issues, clearance issues, or both, the main bearing capscrews **must** be tightened within 15 minutes of the Loctite™ 518 thread sealant application.



Make sure the bearing shell does **not** move.

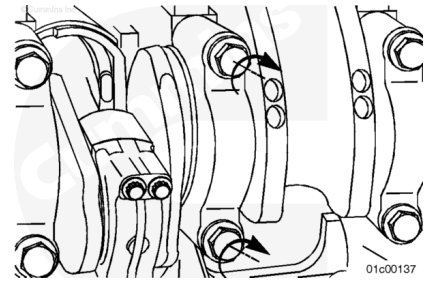
Tighten the capscrews in far enough to use a heel bar to pry the caps into place.



Tighten both of the capscrews on a main bearing cap to 203 N•m [150 ft-lb].

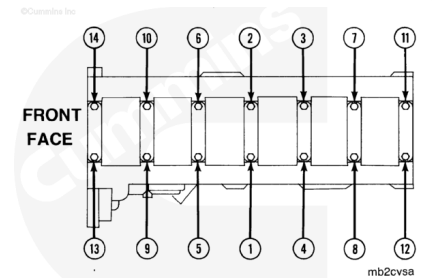
Loosen both capscrews completely.

The remaining main bearings do **not** have thrust bearings.



It is recommended to use an air impact wrench and socket, Tool Numbers 3163059 and 5573072, to rotate the capscrews an additional 180 degrees.

Use the torque plus angle method to tighten the main bearing capscrews in the sequence shown.



Torque Value:

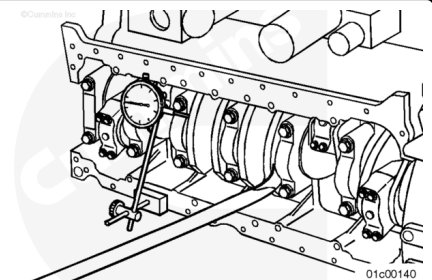
1. 149 n•m [110 ft-lb]

2. Rotate 180 degrees.

To measure the end clearance of the crankshaft, install a dial indicator onto the oil pan flange.

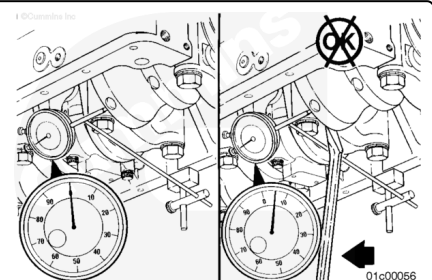
Place the tip of the gauge against the crankshaft counterweight.

Push the crankshaft toward the rear of the cylinder block.



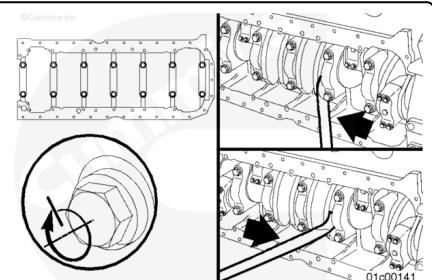
Set the dial indicator at "0" (zero).

Push the crankshaft toward the front of the cylinder block.



If the end clearance is less than 0.10 mm [0.004 in], loosen all main bearing capscrews one turn.

Push the crankshaft toward the front and then toward the rear of the cylinder block.



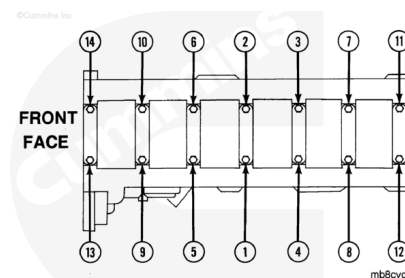
Tighten the main bearing capscrews in sequence.

Torque Value:

Block Main Bearing Capscrews

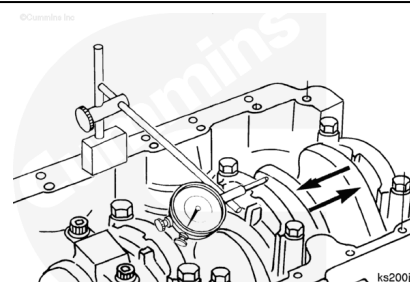
1. 149 n•m [110 ft-lb]

2. Rotate 180 degrees.



Measure the crankshaft end clearance.

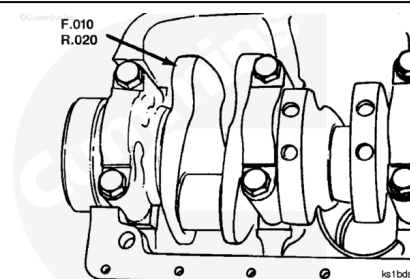
Crankshaft End Clearance - New or Reground		
mm		in
0.10	MIN	0.004
0.50	MAX	0.020



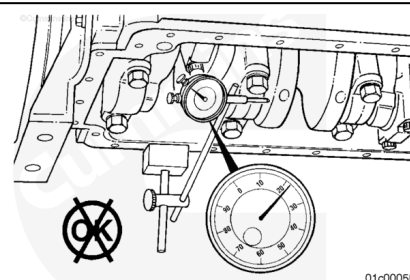
Crankshafts that have been reground on the thrust bearing surfaces are marked for oversize thrust bearings on the rear crankshaft counterweight. If the crankshaft counterweight is marked, check the thrust ring part number to make sure the correct thrust ring size is used.

Example: F0.010 = front 0.25 mm [0.010 in]

Example: R0.020 = rear 0.51 mm [0.020 in]



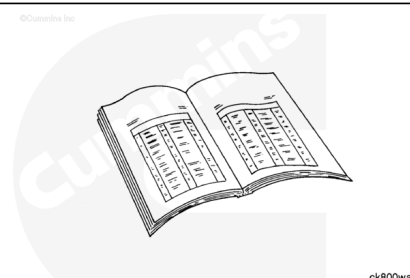
If the crankshaft end play is more than 0.50 mm [0.020 in], the engine **must** be removed to allow for replacement of the crankshaft. Refer to Procedure 000-001 (Engine Removal) in Section 0. (/qs3/pubsys2/xml/en/procedures/10/10-000-001.html)



Finishing Steps

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after any internal engine repairs to avoid internal component damage.



- Install the block stiffener plate. Refer to Procedure 001-089 (Block Stiffener Plate) in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)
- Install the lubricating oil pump. Refer to Procedure 007-031 (Lubricating Oil Pump) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 (Lubricating Oil Suction Tube (Block Mounted)) in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 (Lubricating Oil Pan) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Prime the lubricating system. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Operate the engine until it reaches a temperature of 82°C [180°F], and check for lubricating oil leaks.